CHAPTER 6

# CELL CYCLE AND CELL DIVISION

**CHAPTER ESSENCE AT A GLANCE:**
Sequential cell cycle phases coordinate genomic replication during interphase and equitable chromosome distribution during mitosis and meiosis.

## 4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Interphase Cell Cycle Kinetics:** Interphase occupies 95% of cell cycle duration, partitioned into G1 phase (active metabolic growth, organelle replication), S phase (nuclear DNA replication where DNA content doubles from 2C to 4C while chromosome number remains constant at 2n; centriole duplicates in cytoplasm), and G2 phase (synthesis of tubulin spindle proteins in preparation for mitosis).

• **Mitosis Equational Partitioning:** Mitosis preserves chromosomal ploidy (equational division) across four stages: Prophase (chromatin condensation into visible twin chromatids), Metaphase (bipolar spindle fibers attach to kinetochores and align chromosomes at the equatorial metaphase plate), Anaphase (centromeres split, sister chromatids migrate to opposite poles), and Telophase (nuclear envelopes reform).

• **Meiosis I Reduction and Prophase I Substages:** Meiosis generates haploid gametes through two nuclear divisions. Prophase I features five substages: Leptotene (compaction), Zygotene (synapsis of homologous chromosomes forming synaptonemal complex and bivalents), Pachytene (crossing over mediated by recombinase), Diplotene (dissolution of synaptonemal complex revealing X-shaped chiasmata), and Diakinesis (chiasmata terminalization).

• **Cytokinesis Divergence in Plants and Animals:** Cytokinesis partitions the cytoplasm: animal cells utilize a peripheral contractile actin-myosin ring forming a cleavage furrow that deepens centripetally; plant cells synthesize a rigid cell plate developing centrifugally from Golgi-derived phragmoplast vesicles outwards to fuse with lateral walls.

## ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\text{S-Phase Genome Kinetics} = 2n\ (2C) \xrightarrow{\text{DNA Replication}} 2n\ (4C) \quad [\text{Chromosomes constant, DNA doubled}]$$

*Chromosomal Number and Genomic DNA Mass during Interphase*

$$\text{Prophase I Substages} = \text{Leptotene} \rightarrow \text{Zygotene (Synapsis)} \rightarrow \text{Pachytene (Crossing Over)} \rightarrow \text{Diplotene} \rightarrow \text{Diakinesis}$$

*Sequential Chromosomal Events in Meiotic Prophase I*

$$\text{Anaphase Separation} = \frac{\text{Mitosis: Centromeres split (Sister Chromatids separate)}}{\text{Meiosis I: Centromeres intact (Homologous pairs separate)}}$$

*Centromere Behavior and Chromosome Segregation*

$$\text{Cytokinesis Vectors} = \frac{\text{Animal Cleavage Furrow: Centripetal (Periphery to Center)}}{\text{Plant Cell Plate: Centrifugal (Center to Periphery)}}$$

*Cytoplasmic Partitioning Vectors in Eukaryotes*

## CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In S phase of the cell cycle, DNA amount doubles from 2C to 4C, but the chromosome number remains strictly 2n with no change in ploidy.

• **Alert:** In Anaphase of mitosis, centromeres split and sister chromatids separate; in Anaphase I of meiosis, centromeres do NOT split and homologous pairs separate.

• **Alert:** Crossing over occurs between non-sister chromatids of homologous chromosomes during pachytene of prophase I, facilitated by recombinase enzyme.

## HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

**1. If a diploid onion root tip cell has 16 chromosomes at G1 phase, calculate the chromosome number and DNA content at G2 phase and at Anaphase of mitosis. [Score: 2 Marks]**

**Answer & Key Valuation Steps:**

• At G2 phase: Chromosome number remains 16 (2n), while DNA content doubles from 2C to 4C. **[1 Mark]**

• At Anaphase of mitosis: Centromeres split, temporarily doubling the chromosome count to 32 chromosomes (sister chromatids moving apart) with 4C DNA content. **[1 Mark]**

**2. Name the specific substages of Prophase I in Meiosis characterized by: (a) Formation of the synaptonemal complex, (b) Crossing over between non-sister chromatids, (c) Appearance of chiasmata, and (d) Terminalization of chiasmata. [Score: 2 Marks]**

**Answer & Key Valuation Steps:**

• (a) Formation of synaptonemal complex: Zygotene stage. **[0.5 Mark]**

• (b) Crossing over between non-sister chromatids: Pachytene stage. **[0.5 Mark]**

• (c) Appearance of X-shaped chiasmata: Diplotene stage. **[0.5 Mark]**

• (d) Complete terminalization of chiasmata: Diakinesis stage. **[0.5 Mark]**

**3. Describe the morphological and chromosomal changes occurring across the five sequential substages of Prophase I in Meiosis. [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• Leptotene: Chromosomes gradually condense and become visible as slender long threads under the light microscope. **[0.6 Mark]**

• Zygotene: Homologous chromosomes begin pairing together (synapsis) accompanied by the formation of a complex nucleoprotein structure called the synaptonemal complex, forming bivalents. **[0.6 Mark]**

• Pachytene: Four chromatids of each bivalent become distinct as tetrads; non-sister chromatids undergo crossing over catalyzed by recombinase enzyme, exchanging genetic material. **[0.6 Mark]**

• Diplotene: Dissolution of the synaptonemal complex causes homologous chromosomes to separate, except at sites of crossovers where X-shaped structures called chiasmata are visible. **[0.6 Mark]**

• Diakinesis: Terminalization of chiasmata takes place, chromosomes fully condense, meiotic spindle assembles, and nucleolus and nuclear envelope disintegrate. **[0.6 Mark]**

**4. Differentiate between Anaphase of Mitosis and Anaphase I of Meiosis with respect to centromere behavior, chromatid separation, and ploidy consequences. [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• Centromere behavior: In mitotic anaphase, each centromere splits longitudinally; in anaphase I of meiosis, centromeres do not divide and remain intact. **[1 Mark]**

• Chromatid separation: In mitotic anaphase, sister chromatids separate and move to opposite poles; in meiotic anaphase I, homologous chromosome pairs separate while sister chromatids remain united at centromeres. **[1 Mark]**

• Ploidy consequences: Mitotic anaphase preserves diploid (2n) state equational division; meiotic anaphase I reduces chromosome number by half from 2n to n (reductional division). **[1 Mark]**

**5. Compare cytokinesis in an animal cell with that in a plant cell, describing the roles of the cleavage furrow and cell plate. [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• Animal cell cytokinesis: Accomplished by the appearance of a furrow in the plasma membrane that deepens centripetally (from outside inwards) until it meets in the center and cleaves the cell into two. **[1.5 Marks]**

• Plant cell cytokinesis: Inextensible rigid cell walls prevent furrowing; instead, a new cell wall begins forming in the center as a cell plate (phragmoplast), growing centrifugally (from center outwards) to join preexisting lateral walls. **[1.5 Marks]**

**6. Explain the biological significance of meiosis in sexually reproducing organisms. [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• Conservation of chromosome number: Reduces chromosome number by half (from diploid 2n to haploid n) in gametes, ensuring that fertilization restores the species-specific chromosome number across successive generations. **[1.5 Marks]**

• Genetic variability: Crossing over during pachytene and independent assortment of maternal and paternal chromosomes generate novel genetic combinations, which provide raw material for natural selection and evolution. **[1.5 Marks]**

**7. A plant tissue culture is treated with colchicine, a chemical that selectively inhibits the polymerization of tubulin microtubules into spindle fibers. Predict: (a) The stage at which mitosis arrests, (b) The state of chromosomes, (c) The effect on cytokinesis, and (d) The ploidy level of the resulting cells. [Score: 4 Marks]**

**Answer & Key Valuation Steps:**

• (a) Mitotic arrest stage: Cells arrest at Metaphase because kinetochores cannot attach to functional spindle fibers to guide equatorial alignment. **[1 Mark]**

• (b) Chromosomal state: Chromosomes remain condensed and duplicated as paired sister chromatids, unable to migrate to opposite poles. **[1 Mark]**

• (c) Effect on cytokinesis: Cytokinesis is completely blocked because no telophase nuclei or phragmoplast vesicles form. **[1 Mark]**

• (d) Resulting ploidy: Nuclear envelope reforms around all duplicated chromosomes without division, doubling ploidy from diploid (2n) to tetraploid (4n), inducing polyploidy. **[1 Mark]**